Hot Wings: Quantifying the Impact of Large Appendages on the Heat Loss Potential and Migratory Decision-making of Humpback Whales

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Abstract

For aquatic endotherms, especially those that migrate between warm and cold waters, efficient thermoregulation is a physiological necessity. Marine mammals rely on the interplay of an insulative blubber layer and countercurrent heat exchange through uninsulated thermal windows (e.g., flippers, flukes) to regulate metabolic heat and maintain a consistent core body temperature. Modeling heat loss potential with and without the contribution of these thermal windows can provide insight into the metabolic scope of an individual (i.e., the minimum and maximum sustainable metabolic rate) and

physiological plasticity. Compared to other cetacean species, humpback whales (*Megaptera novaeangliae*) have uniquely large pectoral flippers and tail flukes relative to their body size. These appendages likely serve as potent thermoregulatory windows, allowing for increased heat loss in comparison to other large cetaceans. In this study, we estimated this heat loss potential for humpback whales (n = 5929 individuals) spanning a wide range of body sizes (3.75-14.8 m) and demographic classes (calves and both lactating and non-lactating adults) in both a warm (Hawai'i) and cold water (Southeast Alaska) environment. Mean daily heat loss of humpback whales was higher in Hawai'i with the thermal windows "open" (5803±1432 MJ day^{-1}) than in Southeast Alaska with the thermal windows "closed" (3158±994 MJ day^{-1}), representing the upper and lower bounds of heat dissipation and conservation, respectively. We also found that adults in Hawai'i are capable of producing metabolic heat equal to or exceeding their maximal heat loss potential. Finally, our model predicted that calves born in Southeast Alaska would require higher minimum metabolic rates to match heat loss than those born in Hawai'i. Depending on the interpretation of heat loss as a distinct energetic cost as opposed to a metabolic byproduct, calves in Southeast Alaska would face an additional total heat loss burden of 6938 MJ (equivalent to ~9.4-17.6% of somatic growth costs during the first 90 days of life). Our study demonstrates that the large appendages of humpback whales provide a powerful mechanism for regulating heat loss, allowing individuals to avoid both hyperthermia and hypothermia across contrasting thermal environments.

Numbered Abstract

1. Efficient thermoregulation is essential for aquatic endotherms, particularly migratory species that move between warm breeding grounds and cold foraging habitats. These species rely on the interplay of an insulative blubber layer and countercurrent heat exchange through uninsulated thermal windows (e.g., flippers, flukes) to regulate metabolic heat and maintain a consistent core body temperature.

2. Compared to other cetacean species, humpback whales (*Megaptera novaeangliae*) have uniquely large pectoral flippers and tail flukes relative to their body size. These appendages likely serve as potent thermoregulatory windows, allowing for increased heat loss in comparison to other large cetaceans. However, the contribution of these appendages to whole-body heat loss across body sizes, life stages, and environments has not been quantified.

3. In this study, we estimated the heat loss potential of the body and appendages for humpback whales (n = 5929 individuals) spanning a wide range of body sizes (3.75-14.8 m) and demographic classes (calves and both lactating and non-lactating adults) in both a warm (Hawai'i) and cold water (Southeast Alaska) environment.

4. Mean daily heat loss of humpback whales was higher in Hawai'i with the thermal windows "open" (5803±1432 MJ day^{-1}) than in Southeast Alaska with the thermal windows "closed" (3158±994 MJ day^{-1}), representing the upper and lower bounds of heat dissipation

and conservation, respectively. We also found that adults in Hawai‘i are capable of producing metabolic heat equal to or exceeding their maximal heat loss potential. Finally, our model predicted that calves born in Southeast Alaska would require higher minimum metabolic rates to match heat loss than those born in Hawai‘i.

5. Depending on the interpretation of heat loss as a distinct energetic cost as opposed to a metabolic byproduct, calves in Southeast Alaska would face an additional total heat loss burden of 6938 MJ (equivalent to ~9.4-17.6% of somatic growth costs during the first 90 days of life).

6. Our study demonstrates that the large appendages of humpback whales provide a powerful mechanism for regulating heat loss, allowing individuals to avoid both hyperthermia and hypothermia across contrasting thermal environments.

Introduction

Maintaining a favorable balance between energetic gain and expenditure is critical for the continuation of life. For animals, energetic intake comes from a singular source (foraging), but energetic costs are often varied and include complex behavioral and physiological processes such as locomotion (for motile species) (Alexander, 2005; Schmidt-Nielsen, 1972), prey capture (Bolnick and Ferry-Graham, 2002; Griffiths, 1980), digestion (Armstrong and Blaxter, 1956), somatic growth (Christiansen et al., 2022; Millward et al., 1976; van Aswegen et al., 2025c), reproduction (Bell, 1980; Remick, 1992), thermoregulation (Scholander et al., 1950), and basal metabolism (He et al., 2023; Kleiber, 1975; Spaargaren, 1994). For species that can be housed under human care, these costs can oftentimes be quantified holistically or separated out into their component processes (Allen et al., 2022; Fahlman et al., 2025; Lindström et al., 1999; Williams and Davis, 2024). For larger or cryptic species that cannot be housed under human care, or for situations where species cannot be reliably observed without significant changes to behavior, energetic estimates are often modeled through a patchwork of quantifiable parameters and up-to-date approximations from more tractable species or historical records (Boyles and Brack, 2009; Gough et al., 2022; Pirotta et al., 2018).

The infraorder *Cetacea* includes many large-bodied, highly mobile species that cannot be ethically or logistically maintained in human care. Field experiments on large, free-swimming cetaceans require ideal conditions, large swaths of open ocean, and cutting-edge technology (e.g., biologging tags, unmanned aerial systems [UAS] drones, acoustic monitors) to interpret changes in behavior (Castellote et al., 2012; Curé et al., 2013; Curé et al., 2015; Southall et al., 2016). These constraints have made large cetaceans strong candidates for computational modeling, both to generate energetic baselines (Christiansen et al., 2023; Fahlman et al., 2025; Goldbogen et al., 2019a; Gough et al., 2022; Gough et al., 2025) and to explore the potential physiological consequences and energetic costs associated with disturbance (Christiansen et al., 2025; Czapanskiy et al., 2021; Hin et al., 2019; Pirotta et al., 2018).

Long-distance migrations between cold foraging areas and warm breeding habitats add significant complexity to energetic modeling of large whales. Understanding the annual energetic cycle of these species requires empirical data or accurate estimations of various costs for both sides of the migratory route. Given the substantial differences in water temperature between the two locations (as much as ~30°C), thermoregulation is likely among the most energetically costly processes affected by migration, requiring a suite of adaptations to either conserve or dissipate heat in response to environmental conditions (Favilla et al., 2024). Researchers have posited multiple behavioral reasons for the migratory life history of large whales, including predator avoidance in the early stages of life, enhanced parasite removal, seasonality of resources, or the possibility of a more favorable thermal environment during early development (Clapham, 2001; Corkeron and Connor, 1999; Pitman et al., 2020). This latter hypothesis treats thermoregulation as a distinct energetic cost. However, recent studies (Christiansen et al., 2026; Sumich, 2021) have challenged this interpretation, suggesting instead that heat loss is simply a byproduct of metabolism. These studies have found the heat loss of newborn calves to be lower than their estimated FMR, implying that thermoregulation is unlikely to drive migration provided the calves can adequately dissipate excess metabolic heat (Christiansen et al., 2026; Sumich, 2021).

The majority of cetacean heat loss is thought to be modulated through regional heterothermy in the extremities (i.e., flippers, tail flukes, and dorsal fin), with a counter-current heat exchange system that either restricts or facilitates the movement of warm blood near to the skin's surface (Scholander and Schevill, 1955). This system consists of rosettes of rings of veins surrounding a central artery that permits heat exchange to route warm blood back to the body. When a large heat load must be dissipated, a rise in blood pressure increases the diameter of the central artery collapsing the surrounding veins and distributing blood to superficial veins beneath the skin. Body and appendage sizes should, therefore, have a significant impact on thermoregulation. Bergmann's rule suggests that larger animals with lower surface-to-volume ratios will inhabit colder environments at higher latitudes, while Allen's rule suggests that species living in colder environments will have smaller appendages to decrease excess heat loss (Ray, 1960; van Aswegen et al., 2019). Among the large whales, the humpback whale (*Megaptera novaeangliae*) exist at an intersection of these two rules, having large body sizes (~30,000 kg; ~12-14 m in length; van Aswegen et al., 2025c), but also disproportionately large flippers (~30% of body length) (Fish and Battle, 1995; Kahane-Rapport and Goldbogen, 2018). It is also one of the few large whale species that migrates all the way from near-freezing polar waters to warm tropical waters and back, thereby experiencing an exceptionally wide range of thermal environments.

Models of thermoregulatory costs in large whales have traditionally focused on the heat loss potential of the body, with thermal windows represented as areas of minimal heat loss that remain "closed" (Christiansen et al., 2026; Glarou et al., 2025a; Sumich, 2021). This approach provides a lower bound for heat loss but does not account for the ability of these thermal windows to dissipate excess heat and maintain thermoneutrality. Conversely,

126 estimating the maximal heat loss from these thermal windows establishes an upper limit
127 of heat loss and enables the estimation of the maximal sustainable metabolic rate. For
128 species with large thermal windows, such as humpback whales, this increased surface
129 area for heat dissipation may support higher metabolic rates, and therefore higher activity
130 levels, compared to similarly sized species with smaller appendages.

131 In this study, we combined morphometric data from UAS (unoccupied aerial systems i.e.,
132 drones) with behavioral data from biologging tags (e.g., ventilation rates) and anatomical
133 measurements from stranded animals (e.g., body and appendage blubber thicknesses)
134 into a model of whole-body heat loss potential for humpback whale adults and calves in
135 both a warm-water breeding ground (Hawai‘i) and a cold-water foraging ground
136 (Southeastern Alaska) (Figure 1A). These estimates incorporate context-dependent heat
137 loss through the appendages (Figure 1B-C). Our results indicate that adults possess
138 adequate blubber insulation to tolerate cold summer feeding temperatures (2–5 °C), but
139 may need to reduce activity on warm winter breeding grounds to prevent overheating, even
140 with enhanced heat-dissipating appendages. We also found that the small average body
141 size of newborn calves would necessitate a higher minimum metabolic rate in cold
142 Alaskan waters, but not in warmer Hawaiian waters. This increased minimum metabolic
143 rate required by calves in Southeast Alaska would amount to an additional total heat loss
144 burden of 6938 MJ, if we assume that heat loss is a distinct energetic cost to the animal.
145 This would be equivalent to ~9.43-17.6% of their estimated growth costs during the first 90
146 days of life (van Aswegen et al., 2025c).

Methods

Morphometric Variables From UAS-Photogrammetry

147 For the majority of our analyses, we used a dataset of morphometrics (n = 5929
148 individuals) collected by UAS-photogrammetry on humpback whales between 2018 and
149 2025 across Hawaiian (breeding ground; n = 3701) and Southeast Alaskan (foraging
150 ground; n = 2228) waters. Additional details on the UAS protocols and data collection
151 methodology pertaining to this dataset have been provided by van Aswegen et al. (2025b;
152 2025a; 2025c).

153 Morphometric measurements for each animal included the total body length from the
154 anterior point of the rostrum to the tail notch (TL; in m) and 19 distinct width
155 measurements taken at 5% intervals along the length of the animal. From the combination
156 of these length and width measurements, we estimated the body volume (BV; in m^3)
157 following the approach of Christiansen et al. (2019), applying body height-width ratios
158 measured in van Aswegen et al. (2025c). The body condition of each animal was estimated
159 using residuals calculated from the log-log linear relationship between BV and TL, fitted
160 separately for adult female, male, and unknown-sex whales (van Aswegen et al. 2025c).
161 We estimated total body mass (M_{body}; in kg) following Christiansen et al. (2022), by
162 partitioning the body volume of an animal into tissue types that scale with TL - muscle,

bone, viscera - and blubber volume, which varies with body condition. Body surface area (S_{body}; in m^2) was calculated using the approach of Glarou et al. (2025a), using our estimates of TL, body width, and body height. Calf age was estimated following van Aswegen et al (2025c), by modeling birth size and postnatal growth rates as functions of maternal length. Approximate birth dates were inferred from the predicted body volume at birth and daily growth rates derived from repeated measurements or from the relationship between maternal length and calf body volume growth rates. Figure S1 provides a visualization for a subset of relevant morphometric measurements.

We used a subset of UAS images to determine the surface area of the pectoral flippers (S_{flippers}; in m^2) and tail flukes (S_{flukes}; in m^2). In all images, animals were at the surface with their flippers and flukes extended and flat. We used ImageJ v1.54g (Schindelin et al., 2012) to scale each image to the known total body length of the animal (TL), then outline each extremity to determine the dorsal area ($S_{flippers}$ and S_{flukes}; in m^2). The left and right sides of each pair of extremities were measured separately, then averaged together to account for measurement error. To determine the overall area of each pair of extremities, the "single side" values for $S_{flippers}$ and S_{flukes} were quadrupled. For each individual in our subset of UAS images, we regressed relevant morphometric values ($S_{flippers}$ and S_{flukes}) against TL (Figure S2) and used those predictive models to estimate the extremity areas for each individual in our broader dataset.

Body and Appendage Blubber Thickness

We estimated the body blubber thickness (BT_{body}, m) as a same-thickness sheet surrounding the entire body (without appendages) using a regression equation taken from Glarou et al. (2025a):

$$BT_{\text{body}} = TL * \left(0.0119345 - (0.0003168 * TL)\right) \quad (1)$$

To determine the blubber thickness of the appendages ($BT_{\text{appendage}}$), we obtained flippers and/or flukes from stranded individuals representing multiple species, including the pygmy killer whale (*Feresa attenuata*), striped dolphin (*Stenella coeruleoalba*), harbor porpoise (*Phocoena phocoena*), and humpback whale (measurements are outlined in Table S1). We divided each appendage at three cross-sectional positions (approximately 25%, 50%, and 75% along the span) and measured the average blubber thickness from 12 positions around the periphery of the chord at each position. To determine a single measure of blubber thickness for each appendage, we averaged our measurements (36 in total) for that appendage. For the harbor porpoise and humpback whales in our dataset, we only had single mid-span cross-sectional images. For individuals where we had a flipper and fluke, blubber thicknesses were found to be similar between the two. As a result, we incorporated measures from flippers and flukes into a single "appendage" measurement. For each individual, we also had an estimate of TL, allowing us to create a regression between $BT_{\text{appendage}}$ and TL and use that to determine the $BT_{\text{appendage}}$ for each individual in our broader morphometric dataset (Figure S3).

Conductive Heat Loss

We used the "flat plate" model first introduced by Blix and Folkow (1992) to determine the heat loss (HL_{body}; in W) from the body of an animal to its environment using the equation:

$$HL_{\text{body}} = \frac{k * S_{\text{body}} * \left(T_{\text{body}} - T_{\text{ambient}}\right)}{BT_{\text{body}}} \quad (2)$$

where k was the conductivity of blubber (W m^{-1} °C^{-1}), T_{body} and $T_{ambient}$ were the internal body temperature of the animal and the ambient temperature of the water, respectively (°C). Our model assumed that there was sufficient forced convection to create a negligible boundary layer and maintain the full temperature gradient between T_{body} and $T_{ambient}$.

To determine the heat loss potential from the flippers (HL_{flippers}; in W) and flukes (HL_{flukes}; in W) of the animal, we used the same heat loss equation (Equation 1) and substituted in values specific to each extremity:

$$HL_{\text{flippers}} = \frac{k * S_{\text{flippers}} * \left(T_{\text{flippers}} - T_{\text{ambient}}\right)}{BT_{\text{appendage}}} \quad (3)$$

$$HL_{\text{flukes}} = \frac{k * S_{\text{flukes}} * \left(T_{\text{flukes}} - T_{\text{ambient}}\right)}{BT_{\text{appendage}}} \quad (4)$$

Ventilatory Heat Loss

We determined the average ventilation rates r_v (breaths min^{-1}) of humpback mothers (n=20) and calves (n=32) over a period of hours using methods outlined by Blawas et al. (2025) and accelerometry tag data (Customized Animal Tracking Solutions) collected between 2020-2025 in Hawaiian waters from animals in primarily resting and transiting behavioral states (Hollers et al. *in review*). We did not explicitly incorporate variability due to behavioral state for each individual, instead aggregating all breaths throughout a tag deployment into a single averaged mean. Figure S4 shows the relation between r_v and TL. Using this regression, we estimated r_v for each of the individuals in our morphometric dataset, then used those r_v values to estimate the heat lost to warm expired air during ventilation using an equation from Sumich (2021):

$$HL_{\text{vent}} = \frac{r_v}{60} * \left(143.5 - 1.37 * T_{\text{ambient}} - 0.07 * T_{\text{ambient}}^2\right) \quad (5)$$

Overall Heat Loss Potential

For our overall determination of heat loss potential in each environment, we considered two scenarios: one where the thermal windows (i.e., the flippers and flukes) were "open" and actively contributing to heat loss, and one where those same thermal windows were "closed", allowing the animal to conserve heat (Figure 1B-C). In the "open" condition, the temperature of the flippers and flukes ($T_{flippers}$ and T_{flukes}) was set equal to T_{body}, while in

the "closed" condition, the temperature of the extremities was set equal to the $T_{ambient}$, resulting in a heat loss potential of zero (George et al., 2021). In either scenario, our overall heat loss was determined by combining conductive and ventilatory heat loss using the equation:

$$HL = HL_{\text{body}} + HL_{\text{flippers}} + HL_{\text{flukes}} + HL_{\text{vent}} \quad (6)$$

To incorporate a range of possibilities for heat loss due to factors that cannot be directly measured, we performed a Monte-Carlo simulation (50 iterations with replacement) for each individual in our dataset. In each iteration, we randomly chose values from defined ranges for a series of variables. We used relevant body length (TL) regressions to anchor range midpoints for variables such as body mass (M_{body}), surface area (body, S_{body}; flipper, S_{flipper}; fluke, S_{fluke}), body blubber thickness (BT_{body}), and ventilation rate (r_v). For M_{body} and S_{body}, we chose ranges extending ±10% from those midpoints based on work by van Aswegen (2025c) and our own dataset (Figure S2A), respectively. For S_{flipper} and S_{fluke}, those same ranges were extended ±25% from each midpoint, once again based on our own dataset (Figure S2B-C). For BT_{body}, we used a range of ±10% based on estimates of humpback blubber thickness taken from Glarou et al. (2025a). For $BT_{\text{appendage}}$, we used a wider range extending ±50% based on our dataset showing a greater spread in the residuals of the fitted model regression (Figures S3). For r_v, we based our range on the residual standard deviation of the fitted model, equating to ~±30% around the allometric prediction (Figure S4). For k, we chose a fixed midpoint value of 0.2 (Glarou et al., 2025a; Parry, 1949) and a range of ±10%. Finally, we chose fixed midpoint values of 36°C, 5°C, and 25°C for T_{body}, $T_{ambient}$ in Alaska, and $T_{ambient}$ in Hawaii, respectively (Deser et al., 2010; Hokkanen, 1990; Kagawa-Viviani and Giambelluca, 2020). These temperatures were also sampled from a ±10% range.

Field Metabolic Rate

We inferred the field metabolic rate (FMR) from the mean ventilation rate (r_v) for each individual in our morphometric dataset using an equation:

$$FMR = \frac{1000 * \left(V_t * V_c * LC_{total} * r_v * F_{O_2} * E_{O_2} * C_{O_2}\right)}{60} \quad (7)$$

where V_t was the tidal volume (L breath^{-1}), V_c was the vital capacity, LC_{total} was the total lung capacity (L), F_{O_2} was the fraction of oxygen in atmospheric air, E_{O_2} was the oxygen extraction efficiency, and C_{O_2} was the calorific coefficient of oxygen (kJ L^{-1}). This equation has been used in recent studies to estimate metabolic rates in large whales, including the humpback and southern right whale (*Eubalaena australis*) (Christiansen et al., 2023; Christiansen et al., 2025; Videsen et al., 2023), as well as smaller cetaceans such as the short-finned pilot whale (Gough et al., 2025). Given the various sources of uncertainty in this equation, we included these calculations in our Monte-Carlo simulations (similar to Videsen et al., 2023), generating 100 samples (with replacement) for each individual using paramaterized ranges for each variable. These included a range of 0.4-0.8 for V_t (Fahlman

et al., 2016; Wahrenbrock et al., 1974), a range of 0.8-0.9 for V_t (Fahlman et al., 2017; Kooyman, 1973), and a range of 0.3-0.4 for E_{O_2} (Videsen et al., 2023). For each individual, LC_{total} was calculated from body mass using an allometric equation ($0.135 * \left(M_{\text{body}}^{.92}\right)$) (Fahlman et al., 2011; Kooyman, 1973), then varied by ±20% to account for certainty surrounding lung function (Isojunno et al., 2018). Our estimate of r_v was perturbed ±10% from the allometric prediction to reflect individual and contextual variability in breathing patterns. Finally, we set F_{O_2} at 0.2095 and C_{O_2} at 20.1 (Folkow and Blix, 1992).

Critical Temperature

The upper and lower critical temperatures define the limits of a species' thermoneutral zone (i.e., the range of temperatures within which a consistent metabolic rate can be maintained). We were unable to estimate the upper critical temperature in our model, but we followed methods outlined by Watts et al. (1993) to calculate the lower critical temperature (T_{crit}; in °C) as the ambient temperature at which HL is equal to the metabolic heat generated by the body:

$$T_{crit} = T_{body} - \frac{BT_{body} * HL}{k * S_{body}} \quad (8)$$

Using Kleiber's -(Kleiber, 1975) model of mammalian basal metabolic rate (BMR), we estimated the lower critical temperature for five HL "windows closed" scenarios ranging from 1-2x Kleiber BMR for each individual in our dataset. This provided us with the minimum metabolic rate required to avoid hypothermia for animals of different TL in each environment - in comparison to our minimum and maximum allowable metabolic rates (from our estimation of total heat loss potential) and our predicted FMR from r_v.

Statistics

For analyses of calf age (Figure 7), we fitted generalized additive models for each environment using the function *gam* in the R package *mgcv* (Wood, 2011). We modeled the response as a Gaussian-distributed continuous variable with an identity link, and included calf age (days) as a smooth term using a thin-plate regression spline (s(D)). Smoothing parameters were estimated by restricted maximum likelihood (REML). To reflect differences in data density across environments, the basis dimension (m) was set adaptively as a function of the number of unique age values ($m = min\left(12, max\left(4, \lfloor n_{unique}(age)/3 \rfloor\right)\right)$). Model adequacy was evaluated using standard GAM diagnostics, including residual-versus-fitted plots and Q-Q plots, and by checking whether the chosen basis dimension was adequate (k-index). No additional predictors were considered and no stepwise model selection was performed beyond REML-based smoothing selection. To determine the total cost between the two environments for the age range from 0-90 days (the time that mother-calf pairs are assumed to remain on the calving ground), we predicted both environment fits on a common grid and plotted the

point-wise envelope, restricted to ages where both fits exist. Values are presented as mean ± standard deviation, unless otherwise stated.

Results

Morphometrics

Our morphometric dataset (n = 5929 individuals) included measurements on 1159 calves (n = 1033 in Hawai'i; n = 126 in Alaska), 1155 lactating mothers (n = 1013 in Hawai'i; n = 142 in Alaska), 18 juveniles (n = 17 in Hawai'i; n = 1 in Alaska), 21 yearlings in Hawai'i, and 3576 adults of unknown sex (n = 1617 in Hawai'i; n = 1960 in Alaska). Table 1 presents a subset of important morphometric values for each demographic class and geographic location. Our subset of animals with visually flat flippers and flukes (n = 22) were all from Hawai'i and covered a similar TL range as our larger dataset (3.51-12.69 m). Relationships between TL and selected morphometric estimates (S_{body}; S_{flippers}; S_{flukes}; $BT_{\text{appendage}}$, r_v) are shown in the supplementary materials (Figures S2-S4). In our dataset, S_{flippers} and S_{flukes} were found to be 14.6±1.1% and 11.7±0.9% of S_{body}, respectively. The combination of all appendages (S_{flippers} and S_{flukes}) was, therefore, found to be 26.3±2.0% of S_{body} and 20.8±1.3% of total surface area (sum of S_{body}, S_{flippers}, and S_{flukes}).

Heat Loss

Broken down into the components of HL, we found that daily conductive heat loss potential averaged across all values of TL was highest from the flippers (Hawai'i: 2595±606 MJ day^{-1}; Alaska: 7313±1699 MJ day^{-1}) and flukes (Hawai'i: 2088±484 MJ day^{-1}; Alaska: 5885±1355 MJ day^{-1}). The daily conductive heat loss potential of the body was lower (Hawai'i: 1120±354 MJ day^{-1}; Alaska: 3158±994 MJ day^{-1}), and the ventilatory heat loss potential was exceedingly low (Hawai'i: 0.068±0.046 MJ day^{-1}; Alaska: 0.139±0.096 MJ day^{-1}) (Figure S5). Figure 2 shows the relationship between conductive heat loss and TL for both geographic regions, and Figure S5 shows this same relationship between ventilatory heat loss and TL.

Heat flux was higher for the appendages (Hawai'i: 442.7±169.6 W m^2; Alaska: 1248.0±477.0 W m^2) than it was for the body (Hawai'i: 26.1±7.1 W m^2; Alaska: 73.5±19.9 W m^2), across both geographic regions. Calves displayed higher heat flux in both geographic regions (Hawai'i: 39.4±5.3 W m^2; Alaska: 111.0±14.7 W m^2) than either adults of unknown sex (Hawai'i: 23.1±1.5 W m^2; Alaska: 65.0±3.9 W m^2) or lactating females (Hawai'i: 22.1±0.9 W m^2; Alaska: 62.3±2.1 W m^2).

Combining our conductive and ventilatory heat loss components together, we found mean daily heat loss potential values of 5803±1432 MJ day^{-1} and 1120±354 MJ day^{-1} in Hawai'i for the "windows open" and "windows closed" conditions, respectively. In Alaska, we found a value of 16356±4017 MJ day^{-1} for the "windows open" condition and a value of 3158±994 MJ day^{-1} for the "windows closed" condition. Across a range of body sizes, heat loss was

higher in Hawai‘i with the thermal windows open than in Alaska with thermal windows closed (Figure 3), highlighting the contribution of heat loss from the extremities.

We found that calves displayed the highest average mass-specific heat loss potential (Hawai‘i, 1.27 ± 0.44 W kg^{-1}; Alaska, 3.59 ± 1.23 W kg^{-1}), with lactating females (Hawai‘i, 0.23 ± 0.04 W kg^{-1}; Alaska, 0.65 ± 0.10 W kg^{-1}), and adults of unknown sex (Hawai‘i, 0.27 ± 0.07 W kg^{-1}; Alaska, 0.76 ± 0.20 W kg^{-1}) maintaining similar values (Figure 4). We also found that mass-specific heat loss potential decreased with increasing body condition across all demographic classes.

Sustainable Metabolic Range

From our analysis of lower critical temperature (T_{crit}), the majority of calves from Hawai‘i (74%) were estimated to have critical temperatures between 5°C and 25°C in the 1x Kleiber scenario (Figure 5). Subsequent scenarios led to decreasing T_{crit} values, with the 1.75x and 2x Kleiber scenarios resulting in metabolic heat production from the body exceeding heat loss even in the coldest waters for the smallest calves. In all scenarios, T_{crit} for adults fell well below freezing, putting them outside the range of temperatures experienced anywhere on the planet.

Assuming that heat loss must equal metabolic heat production to maintain thermal balance, we found that the range of sustainable metabolic rates between the “windows closed” condition (i.e., lowest heat loss potential) and the “windows open” condition (i.e., close-to-maximal heat loss potential) was much broader in Alaska, and much narrower in Hawai‘i (Figure 6). In Alaska, the smallest calves (TL of <4 m) had a minimum sustainable metabolic rate of 1.6x Kleiber and a maximum sustainable metabolic rate of 10x Kleiber. In Hawai‘i, these same small calves had a minimum sustainable metabolic rate of 0.53x Kleiber and a maximum sustainable metabolic rate of 3.4x Kleiber. At the larger end of the size scale (TL of >12 m), animals in Alaska had a minimum sustainable metabolic rate of 0.50x Kleiber and a maximum sustainable metabolic rate of 2.3x Kleiber. Those same animals in Hawai‘i had a minimum sustainable metabolic rate of 0.17x Kleiber and a maximum sustainable metabolic rate of 0.78x Kleiber.

Figure 7 shows the minimum sustainable metabolic rate for calves in closer detail, relative to estimated calf age. Heat loss potential (“windows closed” condition) in Hawai‘i was lower than it would be in Alaska for each of the first 90 days of life. We found the total difference between the minimum metabolic rates in Alaska and Hawai‘i over the first 90 days of life to be 6938 MJ. If this total can be split evenly between each day, that would result in a daily difference of 76.2 MJ day^{-1}. Somatic growth costs for a humpback calf have been estimated by van Aswegen et al. (2025) for multiple points within that same age range (0 days, 390 MJ day^{-1}; 30 days, 659 MJ day^{-1}; 60 days, 871 MJ day^{-1}; 90 days, 728 MJ day^{-1}). Our daily difference would therefore account for 17.6% of growth costs at 0 days of age, 10.4% at 30 days of age, 7.88% at 60 days of age, and 9.43% at 90 days of age.

Discussion

Thermoregulation is a complex physiological process, with small changes in core body temperatures leading to a host of adverse effects among endotherms (Benarroch, 2007). Due to the greatly increased thermal conductivity of water in comparison to air (~23x), these physiological constraints are even tighter for fully aquatic species (Favilla and Costa, 2020). Among these, cetaceans are unique in that they have an extreme body size range - from <50 kg for a vaquita (*Phocoena sinus*) (Hohn et al., 1996) up to >150,000 kg for a blue whale (*Balaenoptera musculus*) (Lockyer, 1976) - even among individuals of the same species (Kahane-Rapport and Goldbogen, 2018). In addition, cetaceans live in waters around the globe, with some species migrating from exceptionally cold environments (0-5°C) to warm tropical waters (~25-30°C) and back within a single year (Kettemer et al., 2022; Kshatriya and Blake, 1988; Modest et al., 2021; Zerbini et al., 2006). Previous research on the thermoregulation of large migratory cetaceans is still somewhat limited, with the majority of studies focusing on single geographic locations (George, 2009; George et al., 2021) or single demographic classes, such as calves (Sumich, 2021) or adults (Glarou et al., 2025a; Parry, 1949). Watts (1993) and Ryg (1993) incorporated species-level models for larger cetaceans, but the fine-scale morphometric data available to parameterize their models was limited. Christiansen et al. (Christiansen et al., 2026) modelled the heat loss of southern right whales across their full range of body lengths, body conditions, and geographic distribution. However, their model was restricted to conductive heat loss from the body only and did not include heat loss from the flippers or flukes. Our study is the first to explicitly incorporate heat loss from the appendages of the whales, across a wide range of body sizes (from newborn calf to adult) using morphometrics from a large number of animals on both the breeding and feeding grounds.

Model Limitations

In our estimation of heat loss potential from the body, we relied on a simplified view of blubber, with uniform thickness (BT_{body}) and thermal conductivity (k). Both of these factors have been shown to vary due to a number of critical factors (e.g., timing in the annual cycle, location on the body, temperature gradient between the body and ambient environment, amount of blood perfused through the tissue, lipid composition) (Favilla et al., 2024; George, 2009; Lockyer, 1987; Meagher et al., 2008; Miller et al., 2011; Sumich, 2021; Víkingsson, 1990). Additionally, we were able to incorporate the effects of body condition into our estimation of surface area (S_{body}) following the methods of Glarou et al., (2025a), but we were unable to capture how body condition might also impact BT_{body} or k.

In comparison to the body, modeling heat loss potential through the extremities is even more difficult. There are many experimentally-derived values in the literature that we can draw from for the blubber anatomy and thermal dynamics of large whale bodies (Glarou et al., 2025a; Parry, 1949). For the extremities, we relied on the same value of k for both the body and the extremities, and we based our estimation of $BT_{\text{appendage}}$ on the cross-sections of flippers and flukes from multiple species and age/sex classes. These values ranged from

0.47-8.3, though the values from the two ~9 m humpback whales in the dataset were remarkably different (~3 mm in the subadult flipper cross-section versus ~8.3 mm in the juvenile fluke). Unfortunately, there are very few literature sources on the BT of cetacean extremities to compare against. The majority of studies that investigate the flippers or flukes focus instead on other anatomical structures (e.g., skin, collagenous fibers, bone, vasculature), noting only a very thin blubber layer (Felts, 2023; Gavazzi et al., 2024; Gough et al., 2018; Sun et al., 2010). Only George (2009; 2021) gives actual values for the blubber thickness of the flippers (~20 mm) and flukes (~10-15 mm), but these were for the strongly ice-associated bowhead whale (*Balaena mysticetus*) that spends its entire life in cold arctic waters.

Our model assumes the same thermal dynamics in the appendages as the rest of the body, with an internal temperature (set as the same as the core body temperature) and a single-thickness blubber layer that loses heat uniformly to the environment. Heat loss through the cetacean extremities is tied closely to the anatomical layout of their vasculature (Favilla et al., 2024; Williams et al., 1999), suggesting that heat loss in a more detailed model would take into account areas of higher and lower heat loss potential. As a result of this simplification, we could be overestimating the heat loss through the extremities. Alternately, a change in the k of blubber on the extremities could lead to additional heat loss that we are unable to capture. Finally, when working with such thin BT values as we included in our model, small changes can have a large impact on model outputs. Additional data will be needed to parameterize these critical variables in future modeling exercises.

Our model also simplifies the effects of forced convection on heat loss potential, assuming that there is always enough water movement past the skin to nullify the boundary layer and maintain the full predicted temperature gradient between T_{body} and $T_{ambient}$. In reality, warming of the boundary layer is likely to reduce this temperature gradient and, subsequently, the heat loss potential of an animal. For adult humpbacks in Hawai‘i that remain stationary for large portions of their day, this could bring them closer to overheating, further necessitating a reduced metabolic rate or additional heat loss mechanisms (e.g., engulfing cool water to engage the highly vascularized tongue and palate (Ford et al., 2013; Heyning and Mead, 1997) that we are unable to capture with our model.

Impact of Thermal Windows on Conductive Heat Loss

Recent models of heat-loss potential in large cetaceans have excluded the extremities (Christiansen et al., 2026; Glarou et al., 2025a; Sumich, 2021), focusing instead on core body heat loss and assuming that, provided FMR exceeds body heat-loss potential, animals can maintain regional heterothermy or modulate thermal windows as needed to maintain body temperature within a thermoneutral range (Sumich, 2021). We sought to test this assumption by developing a model that includes a simple estimation of heat loss potential through the extremities. With this holistic model, we could estimate the previously defined lower bound of heat loss potential (with the thermal windows “closed”)

as well as the presumed upper bound of heat loss potential (with the thermal windows fully "open"), the latter of which may be defined as the maximum thermoregulatory cost that the animal could sustain without overheating (Hokkanen, 1990).

Previous research by Meagher et al. (2008) described the heat flux through different areas of the body in both summer and winter for bottlenose dolphins (*Tursiops truncatus*) in Florida waters. They found only slightly higher summertime values for the flippers (155.3 ± 15.1 W m^2) and flukes (126.2 ± 12.4 W m^2) relative to the body (109.1 ± 5.3 W m^2). Our equivalent heat flux values for Hawai'i showed a much greater discrepancy between the flippers (442.7 ± 169.6 W m^2) and flukes (442.7 ± 169.6 W m^2) and the rest of the body (26.1 ± 7.1), with heat loss through the extremities accounting for $79.8 \pm 0.52\%$ of total conductive heat loss potential in Hawai'i. This percentage was far greater than the 10-15% predicted by Ryg et al. (1993). Their prediction was based primarily on species with much smaller appendages, relative to body size. For a large cetacean, they predicted the surface area of appendages to be ~10% of total surface area, while our humpbacks displayed appendages that were $26.3 \pm 2.0\%$ of body surface area and $20.8 \pm 1.3\%$ of total surface area.

The uniquely large appendages of humpback whales result in enhanced maneuverability and allow for a wide repertoire of foraging behaviors (e.g., bubble-netting) on the feeding grounds (Fish and Lauder, 2006; Nemeth et al., 2025; Szabo et al., 2024; Woodward et al., 2006), but they likely also convey a higher heat loss potential on the warm breeding grounds, resulting in a greater scope for movement or energetically intense behaviors such as competitive male "heat runs" (Tyack and Whitehead, 1982) or repeated breaches (Segre et al., 2020). Bergmann's rule (that larger animals should inhabit colder waters) and Allen's rule (that animals residing in colder climates have smaller appendages) have been suggested as reasons for the appendage-to-body size ratios in various cetacean species (Adamczak et al., 2020; Favilla et al., 2024; Torres-Romero et al., 2016), but humpbacks appear to defy these rules by migrating between both tropical and arctic climates while having the largest appendages (relative to body size) of any cetacean (Kahane-Rapport and Goldbogen, 2018; Woodward et al., 2006). The large size of humpback whale appendages may therefore be key to their ability to tolerate wide variation in water temperatures without risking either hyperthermia or hypothermia.

The large body sizes of humpback whales, and their subsequently increased surface-to-volume ratio, allow for greater heat loss potential in tropical climates to keep from overheating. If our model predicted similarly low heat flux values for both the body and the extremities, adults would be incapable of dispersing enough heat to keep from overheating in Hawaiian waters. Heat flux estimates from the fore- and hind-flippers of harp seals showed higher values (~100-300 W m^2) at ambient temperatures of 23-24°C, with short-duration increases (up to 20-fold) measured for one or both hind-flippers (Kvadsheim and Folkow, 1997). If the countercurrent heat exchange system in cetaceans works similarly, it suggests that our higher extremity heat flux values are not outside the realm of possibility. It is also possible that whales can increase heat dissipation from the main body by elevating blood flow to the outer blubber layer, thereby increasing k (Kanwisher and

Sundnes, 1966). In this scenario, whales would rely less on their appendages for heat dissipation in warmer waters or during periods of elevated activity.

Effects of Body Size on Metabolic Rate

Estimating metabolic rates in large cetaceans has remained a persistent challenge, but our understanding of this important physiological metric underlies our broader knowledge of species health and resilience. For smaller species that can be housed under human care, researchers have estimated basal metabolic rates ranging from ~1.5-3.5 times the Kleiber (1975) "mouse-to-elephant" allometric curve (He et al., 2023; Noren and Rosen, 2023; Williams, 2022) - much higher than similarly-sized terrestrial counterparts. For larger-bodied species such as the humpback whale, basal metabolic rates cannot be directly measured through any of the typical experimental pathways (e.g., direct respirometry, monitoring of heart rate, ingestion of doubly-labeled water). Instead, we must rely on anatomically-derived models (Lockyer, 1981) and inform them with any physiological data that we can obtain (Goldbogen et al., 2019b).

These models have, so far, produced estimates broadly consistent with those for smaller cetaceans, suggesting a potential allometric relationship and possibly elevated BMR in larger species (Williams, 2022). However, this remains a matter of considerable debate: while some studies report higher BMR in aquatic mammals relative to terrestrial species, others find no such elevation once methodological and analytical issues are accounted for (e.g., Glarou et al., 2025b; He et al., 2023; Lavigne et al., 1986; McNab, 1988; White, 2010). Thus, whether large cetaceans truly exhibit elevated BMR remains unresolved. However, for the purposes of this study, we assume that BMR scales allometrically as predicted by Kleiber.

If we assume that the entire metabolic budget will eventually need to be offloaded as heat, we can use our model of heat loss with the thermal windows "closed" and "open" as proxies for the minimum and maximum sustainable metabolic rates, respectively. FMR values for smaller cetaceans have been estimated at ~3.5 times the Kleiber prediction (Rojano-Doñate et al., 2018), and the FMR of calves and juveniles is thought to be higher than for adults (Innes et al., 1987; Watts et al., 1993). For the animals on the feeding grounds in Alaska, the metabolic "scope" between minimum and maximum heat loss potential ranged from 1.6 to 10 times the Kleiber prediction for the smallest animals (<4 m) and from 0.50 to 2.3 times the Kleiber prediction for the largest animals (>12 m). These values captured the full range of literature predictions at the lower end of the body size scale, suggesting that calves and juveniles could sustain even higher metabolic rates, if necessary. For larger animals, heat loss potential in the "open" condition is below the literature-based FMR prediction of ~3.5. As stated previously, small changes in the parameters of our appendage-based heat loss model could account for these discrepancies. Individuals may also be relying on additional avenues of heat loss. Research on Gray whales (*Eschrichtius robustus*) has suggested that the intake of water into the mouth - with or without prey - may serve a thermoregulatory function, with the highly vascularized upper palate and tongue acting as thermal windows (Ford et al., 2013;

Heyning and Mead, 1997). The implications of this behavior are difficult to incorporate into a heat loss model, but any additional heat loss would serve to increase the maximal sustainable metabolic rate, allowing for more exertion with less chance of overheating. In comparison to Alaska, we found a much smaller range of sustainable metabolic rates for adults in Hawai‘i (from 0.17 to 0.78 times the Kleiber prediction), but these animals are fasting and spending a large portion of their time resting (Bejder et al., 2019; Christiansen et al., 2025), both factors that curtail expenditure and could maintain FMR values close to - or below - basal levels estimated from Kleiber.

Migratory Decision-Making

Corkeron and Conner (1999) summarized multiple hypotheses for the long-distance migrations observed in large cetacean species such as the humpback whale, concluding that predator avoidance (i.e., leaving waters that contain large numbers of killer whales [*Orcinus orca*] capable of preying on calves) was the likeliest reasoning. Clapham (2001) challenged this hypothesis, noting the absence of documented killer whale attacks in major humpback feeding and breeding areas. He acknowledged, however, that predation may occur along migratory routes (Mehta et al., 2007) and that extended residency on the breeding grounds may allow calves to reach sizes that reduce vulnerability to otherwise fatal attacks. The thermoregulatory explanation - that calves require warmer waters on the tropical breeding grounds to maintain thermoneutrality and maximize energetic investment in somatic growth - was discounted by Corkeron and Conner (1999), citing multiple models of calf thermoregulation that have been performed through the years (Kanwisher and Sundnes, 1966; Watts et al., 1993). In their review and in the subsequent Clapham (2001) commentary, it was noted that additional detail on the anatomical structure and thermal dynamics of cetacean blubber would be necessary to find a sufficient and lasting answer to the migration question.

In the years since, many of these variables have been fleshed out for a number of species (see Glarou et al. (2025a) for a comprehensive overview), and additional models of heat loss have been tested (Christiansen et al., 2026; Sumich, 2021), with the majority agreeing with the assertion by Corkeron and Conner (1999) that heat loss was likely not a major factor in migratory decision-making. The study by Sumich (2021), for example, compared heat loss of gray whale calves against estimates of resting metabolic rate and found that, even under conditions of extreme lipid-depletion, metabolic rates were sufficient to cover heat loss. Similarly, Christiansen et al. (2026) showed that the large body size, and consequent low surface-area-to-volume ratio, of southern right whales, even at birth, is sufficient to offset heat loss, such that only a thin layer of blubber is required to keep thermal loss below their FMR. In our dataset, a small calf would fall well outside its thermoneutral zone if its metabolism were near the basal Kleiber (1975) prediction; however, this scenario is unlikely, as FMR values for small calves (<4 m) were estimated to be 4.3 times the Kleiber prediction. Notably, the minimum sustainable metabolic rate for a calf born in Alaska’s colder waters was substantially higher than for a comparably sized calf born in Hawai‘i, differing by 6938 MJ. If heat loss is considered to be a necessary physiological side-effect of metabolic heat production (as described by Sumich et al.

(2021) and others), then that increase in heat loss would be insignificant so long as it remains below the amount of metabolic heat generated by the animal. But if thermoregulation is thought of as a distinct and realized cost to the animal, then increasing the heat losses in Alaska would necessitate either cuts to the energetic budget in other areas (e.g. somatic growth), as has been posited by Clapham (2001), or increases in the overall metabolic rate to offset increased losses.

Interestingly, one of the few large cetacean species that remains in cold waters throughout the year, the bowhead whale (*Balaena mysticetus*), has been shown to have similar calf growth rates to the humpback - both are born at ~4 m in length, and grow to ~8 m within the first year (George, 2009; van Aswegen et al., 2025c) - suggesting that a non-migratory life history strategy does not necessarily result in slower growth during early life. Bowhead whales are also suspected to have lower core body temperatures (~34°C), resulting in a smaller temperature gradient and subsequently reduced heat loss from the animal to its environment. The lack of large appendages, however, likely limits the bowhead from maintaining a more migratory lifestyle and surviving in warmer environments.

Conclusions

Thermoregulation is a critical physiological function for all marine endotherms, but especially so for humpback whales and other large cetacean species that migrate between cold arctic feeding grounds and warm tropic breeding grounds throughout the annual cycle. Building on the large body of physiological and anatomical work performed on small cetaceans species in human care, as well as the models generated for a variety of larger cetaceans species, we sought to combine our uniquely broad morphometric dataset from humpback whales across a range of body sizes in both Hawai‘i and Alaska into a holistic computational model of heat loss potential under two conditions: with thermal windows “open” and “closed”. From this, we hoped to provide a novel estimate of metabolic scope.

Our results demonstrate that the disproportionately large appendages of humpback whales substantially expand their heat-dissipation capacity on warm breeding grounds, leading to a decreased risk of overheating at large body sizes. For calves, our results show that colder environments impose a higher minimum sustainable metabolic rate and increase the total amount of heat loss during early development (6938 MJ), although these costs remains well below estimated FMR values for similarly-sized animals. Whether this excess heat loss constitutes a meaningful energetic burden or simply a byproduct of metabolic heat production remains unanswered. Regardless, our findings highlight thermoregulation as an important axis of energetic scope that interacts with body size, condition, life stage, and habitat. Incorporating full-body heat loss potential - including the contribution of dynamic thermal windows - into bioenergetic models is essential for understanding migratory decision-making, developmental constraints, and the vulnerability of large whales to future ocean warming.

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Competing interests

The authors declare no competing or financial interests.

Author contributions

Conceptualization: W.T.G., L.B.; Data Curation: W.T.G., M.v.A., L.E., A.S., J.J.C., A.H., F.E.F., K.L.W., J.P., N.H.; Formal Analysis: W.T.G., M.v.A., F.E.F., J.P., N.H.; Funding Acquisition: A.S., J.J.C., K.L.W., L.B.; Investigation: W.T.G., M.v.A., L.E., A.S., J.J.C., A.H., F.E.F., K.L.W., J.P., N.H., L.B.; Methodology: W.T.G., M.v.A., A.F., F.C., F.E.F., L.B.; Project Administration: L.B.; Resources: A.S., K.L.W., L.B.; Software: W.T.G., M.v.A., L.E., A.H.; Visualization: W.T.G.; Writing – Original Draft: W.T.G.; Writing – Review & Editing: all coauthors

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Data and resource availability

Data related to the analyses in this study can be found at Data Dryad (DOI: XXXXX) or at https://github.com/wgough/XXXXXXXX. For more information, please contact the authors. All relevant data and details of resources can be found within the article and its supplementary information.

Tables

Table 1. Morphometrics for animals in each geographic location and demographic class.

Demographic Class	Sample Size		Body Length (cm)		Body Mass (kg)		Surface Area (m2)		Body Condition		Body Blubber Thickness (m)	
	Alaska	Hawaii	Alaska	Hawaii	Alaska	Hawaii	Alaska	Hawaii	Alaska	Hawaii	Alaska	Hawaii
Calf	126	1,033	7.96 ± 0.55	5.38 ± 0.64	8501 ± 2018	2475 ± 947	26.39 ± 3.99	11.70 ± 2.87	0.116 ± 0.139	0.001 ± 0.119	0.075 ± 0.004	0.055 ± 0.005
Juvenile	1	17	8.56	8.39 ± 0.81	8409	9388 ± 3312	27.41	28.60 ± 6.31	-0.100	0.032 ± 0.177	0.079	0.078 ± 0.005
Yearling	-	21	-	9.01 ± 0.57	-	14329 ± 2856	-	36.58 ± 4.93	-	0.309 ± 0.115	-	0.082 ± 0.004
Lactating	142	1,013	12.97 ± 0.83	12.58 ± 0.78	29775 ± 5658	30719 ± 6404	63.44 ± 7.97	63.48 ± 8.37	-0.073 ± 0.092	0.046 ± 0.123	0.101 ± 0.003	0.099 ± 0.003
Adult	1,959	1,617	12.00 ± 0.94	11.37 ± 1.11	25771 ± 6411	21977 ± 6505	56.62 ± 9.15	51.01 ± 9.97	0.000 ± 0.122	-0.009 ± 0.119	0.097 ± 0.004	0.094 ± 0.006
Total	2,228	3,701	11.84 ± 1.34	10.00 ± 3.07	25041 ± 7476	18825 ± 12177	55.33 ± 11.48	43.27 ± 21.97	0.002 ± 0.125	0.012 ± 0.124	0.096 ± 0.007	0.085 ± 0.019

Figures

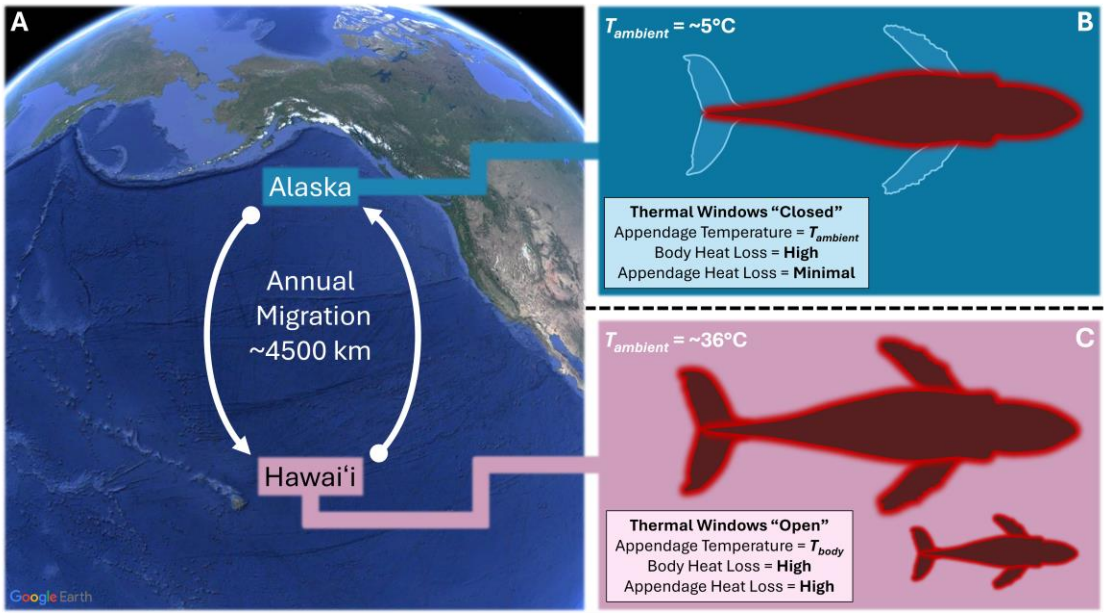


Figure 1: Schematic showing the relationship between geographic location and anatomical heat loss dynamics. A) Map of the North Pacific showing the migratory route for humpback whales between Alaska and Hawai'i. B) the heat loss dynamics for a hypothetical animal in Alaska with thermal windows (i.e., pectoral fins and tail flukes) closed. C) Similar heat loss dynamics for an animal in Hawai'i with thermal windows open and a calf present. Map image sourced from Google Earth version 7.3.6.10441.

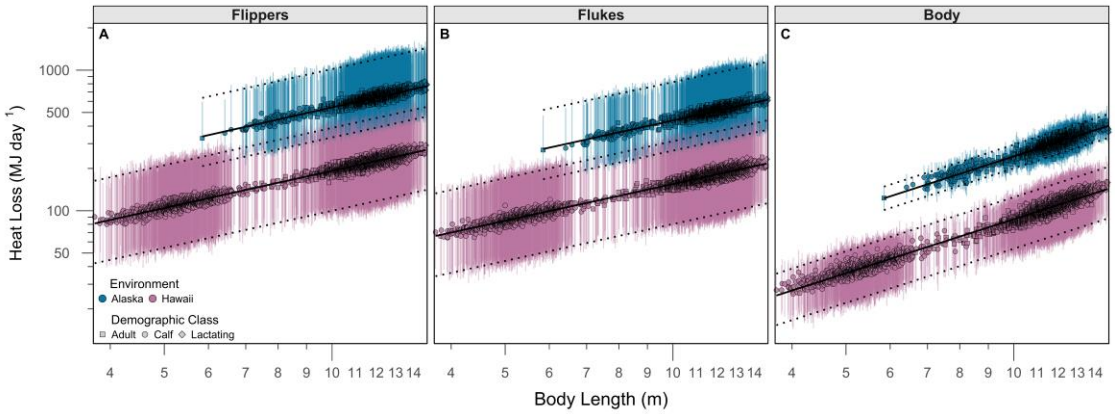


Figure 2: Relationships between body length and potential daily conductive heat loss from A) the flippers, B) the flukes, and C) the body. Data points for the flippers and flukes assume an open thermal state. Points denote median values for each individual, with lines extending along the CI from 5% to 95%, based on our individual-scale Monte Carlo simulations. Dashed and dotted lines denote the regressions for those median and confidence interval values, respectively.

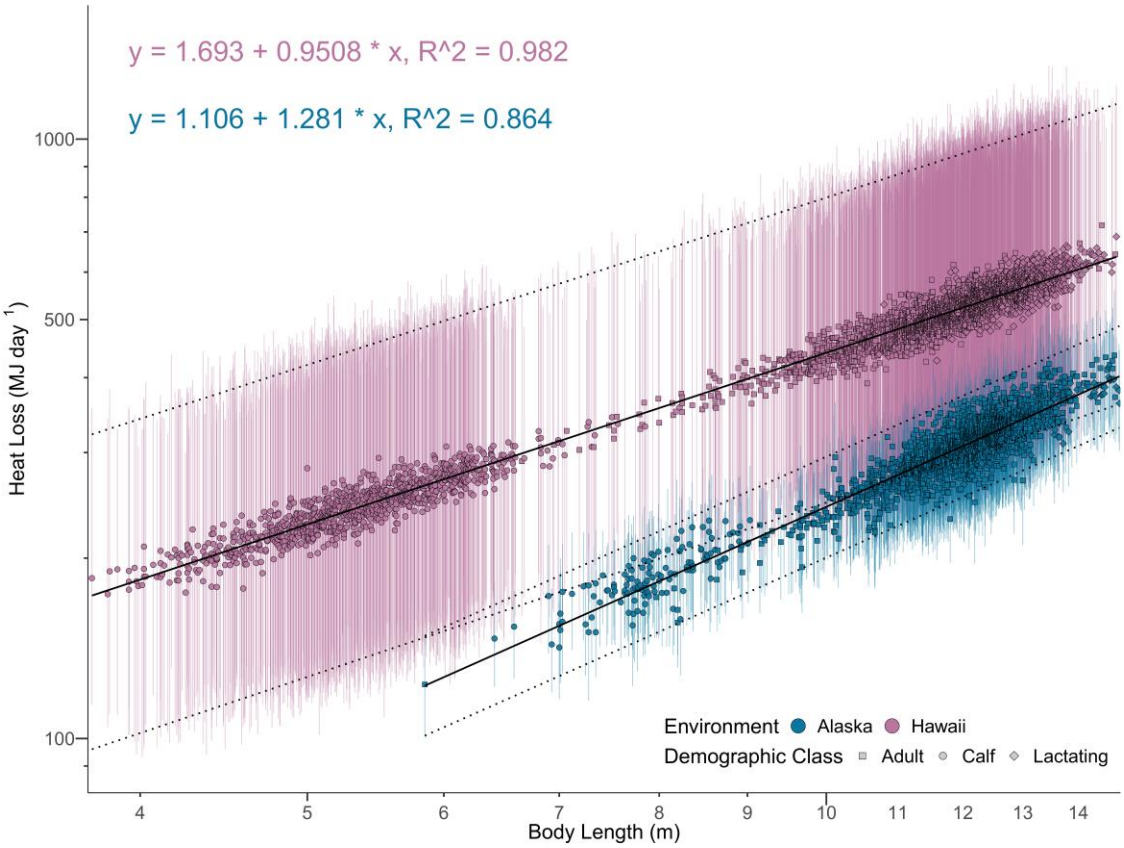


Figure 3: Relationship between body length and potential daily whole-animal heat loss for animals in Alaska (blue) with their thermal windows closed, and for animals in Hawai'i (pink) with their thermal windows open. Points denote median values for each individual, with lines extending along the CI from 5% to 95%, based on our individual-scale Monte Carlo simulations. Dashed and dotted lines denote the regressions for those median and confidence interval values, respectively.

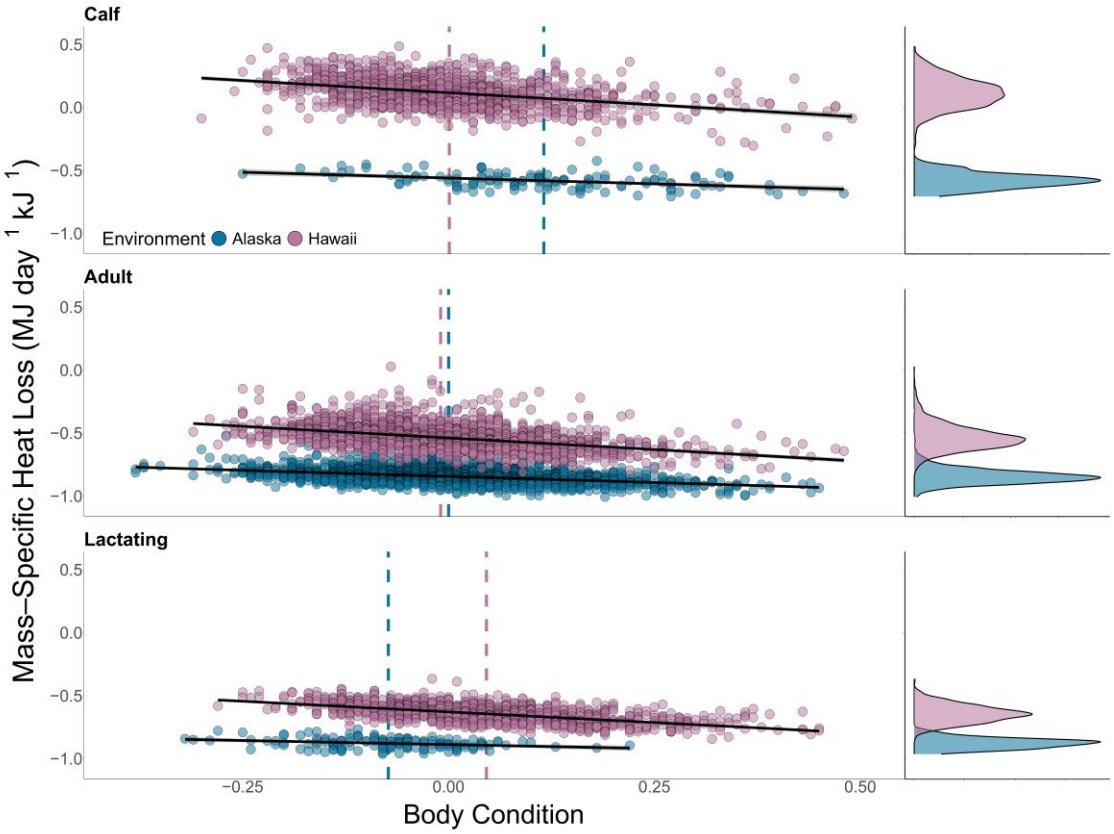


Figure 4: Relationship between body condition and mass-specific heat loss potential (MJ day^{-1} kJ^{-1}) with thermal windows open (Hawai'i; Pink) and closed (Alaska; Blue) for Calves (top), Adults (middle), and Lactating females (bottom). Dotted vertical lines denote the median value for each set of points.

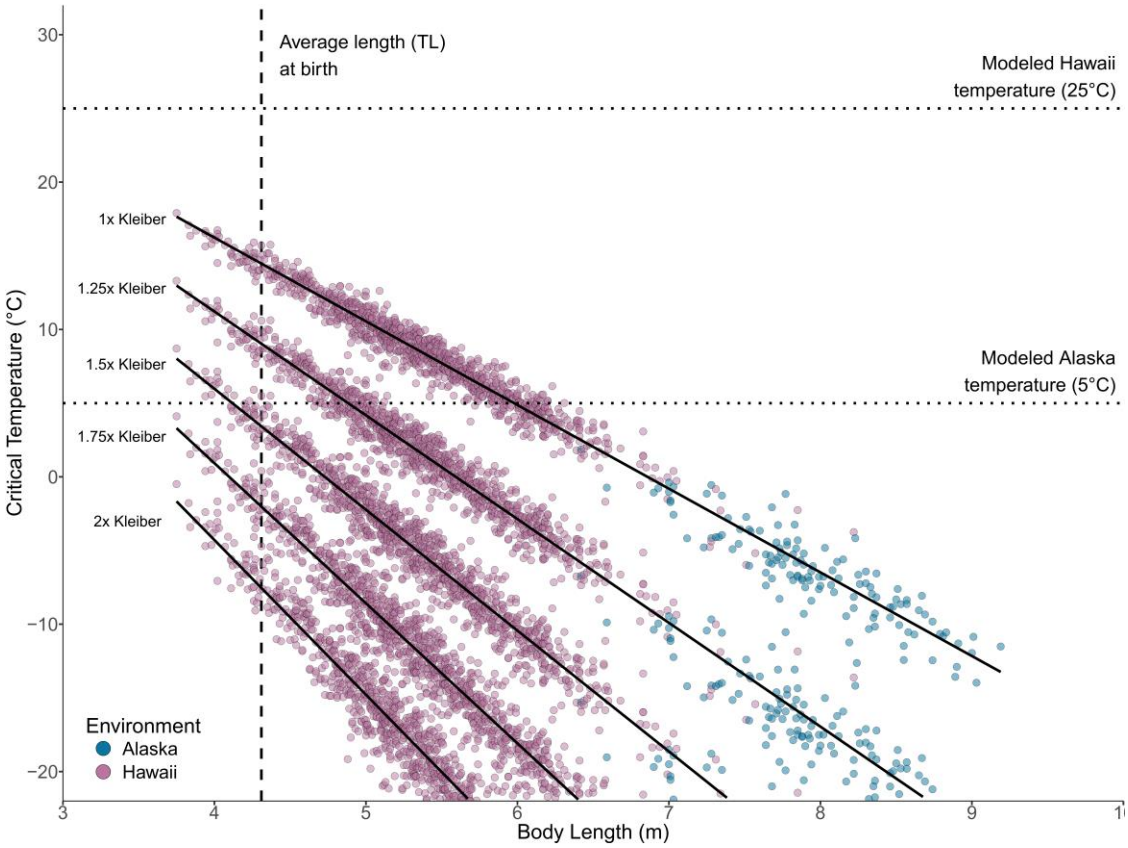


Figure 5: Relationship between body length and lower critical temperature for calves, assuming metabolic heat production values of 1-2x Kleiber BMR (in descending order). Horizontal dotted lines denote the temperatures (°C) used for our warm (Hawai'i) and cold (Alaska) environments. The vertical dotted line denotes the mean estimated TL at birth (4.31 m) for calves in our dataset.

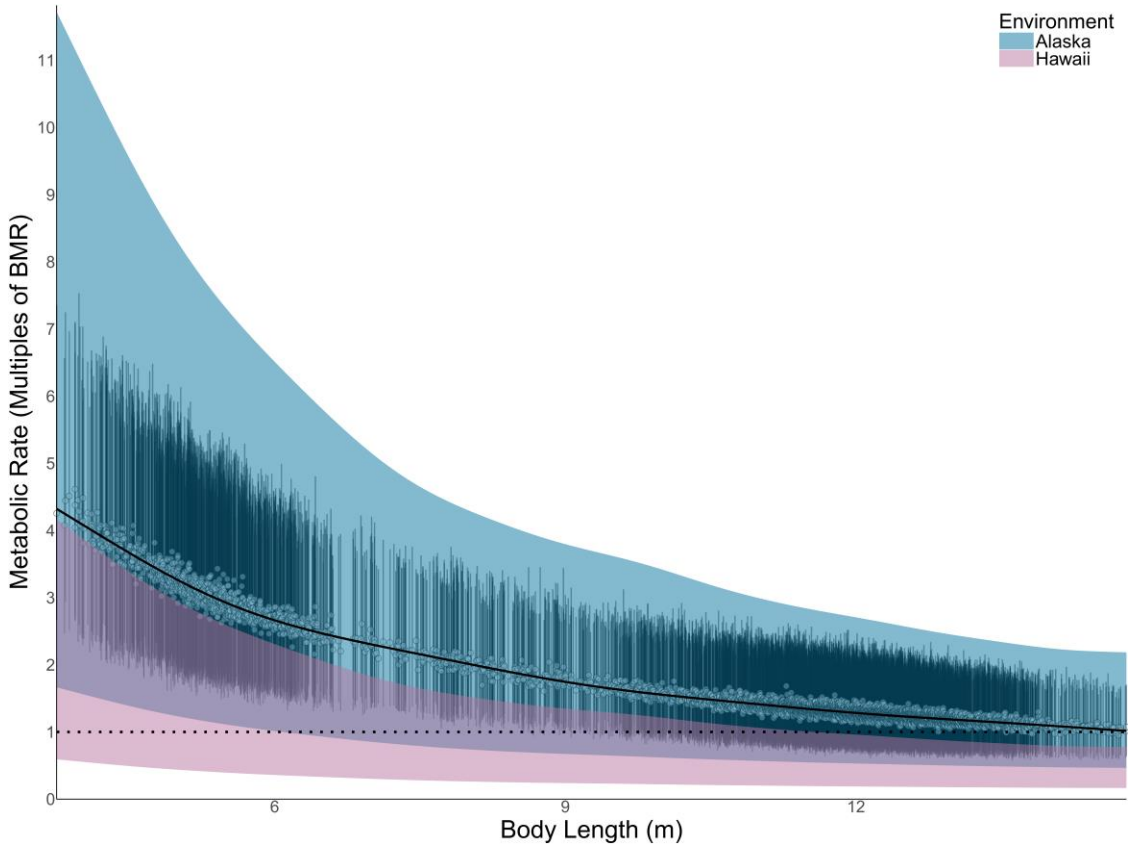


Figure 6: Relationship between body length and the range of potential heat loss values for both Alaska (blue) and Hawai'i (pink), displayed as multiples of "Kleiber" BMR (with the horizontal dashed line denoting 1x Kleiber BMR). The top and bottom of each range correspond to the potential heat loss with thermal windows open and closed, respectively. The darker purple portion is the overlap between the two ranges. Gray points denote median field metabolic rate estimates (derived from our breathing frequency regression) for each individual, with lines extending along the CI from 5% to 95%.

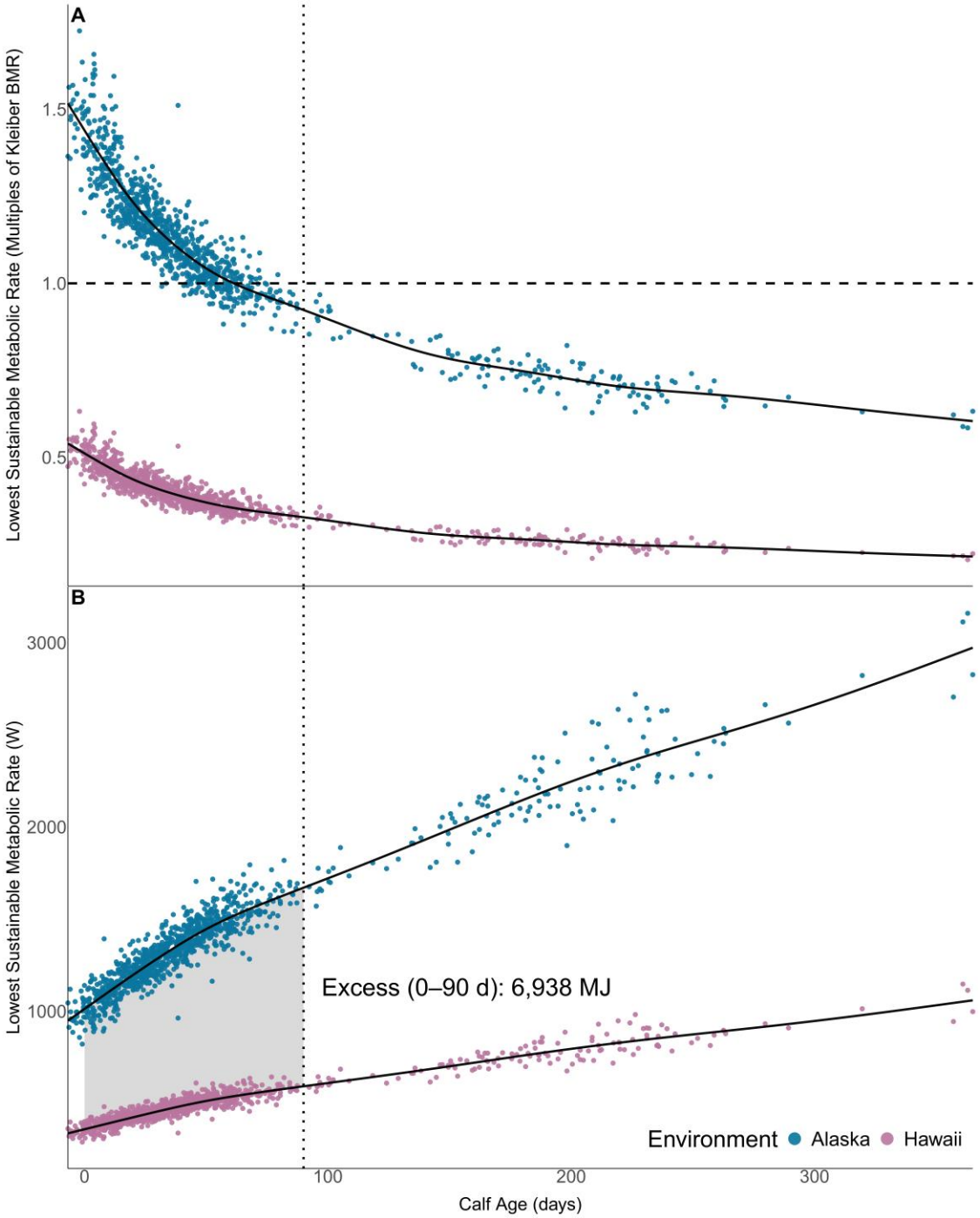


Figure 7: Age vs. the minimum sustainable metabolic rate for calves in Alaska (blue) and Hawai‘i (pink), displayed both as A) multiples of Kleiber BMR or B) absolute wattage. The horizontal dashed line denotes 1x Kleiber BMR, while the vertical dotted line corresponds to 90 days. The area of the shaded region has been calculated to show the cost differences for a calf growing in Hawai‘i vs. Alaska and maintaining its lowest sustainable metabolic rate.

Supplemental Materials

ID	Species	Sex	Age class	Stranding notes	Flipper / fluke	Body length (m)	Appendage blubber thickness (mm)
FA-6436	pygmy killer	Male	Subadult		Flipper	2.13	0.70
FA-9529	pygmy killer	Female	Subadult		Flipper	2.20	0.80
FA-9519	pygmy killer	Female	Adult	Pregnant	Flipper	2.29	1.70
FA-6599	pygmy killer	Female	Adult	Lactating	Flipper	2.32	3.50
FA-8456	pygmy killer	Male	Calf		Flipper	1.37	1.30
FA-2937	pygmy killer	Male	Adult	Emaciated	Flipper	2.32	0.80
FA-7995	pygmy killer	Male	Adult	Emaciated	Flipper	2.35	1.80
SC-3576	striped dolphin	Male	Adult		Flipper/Fluke	2.11	2.05
C-551	humpback whale	Unknown	Juvenile		Fluke	9.26	8.27
FEF-Plasticized	humpback whale	Male	Subadult		Flipper	9.02	3.00
HP2015001	harbor porpoise	Unknown	Unknown		Fluke	1.47	0.47

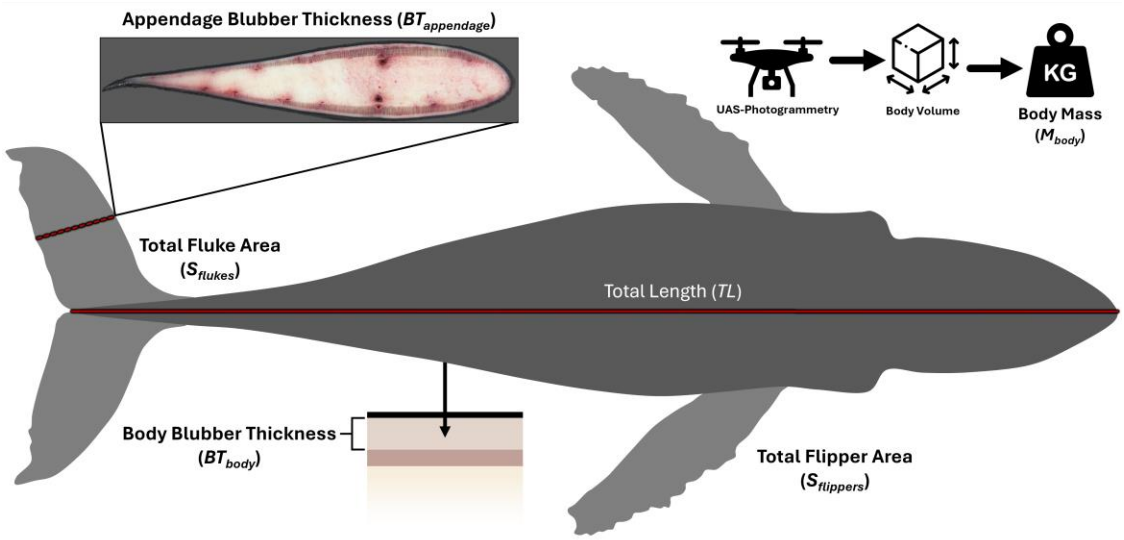


Figure S1: Schematic showing various morphometric and model-critical variables were measured. The cross-section (top left) is from the fluke of a harbor porpoise (C-489), showing the internal layers and vasculature, as well as the fusiform shape.

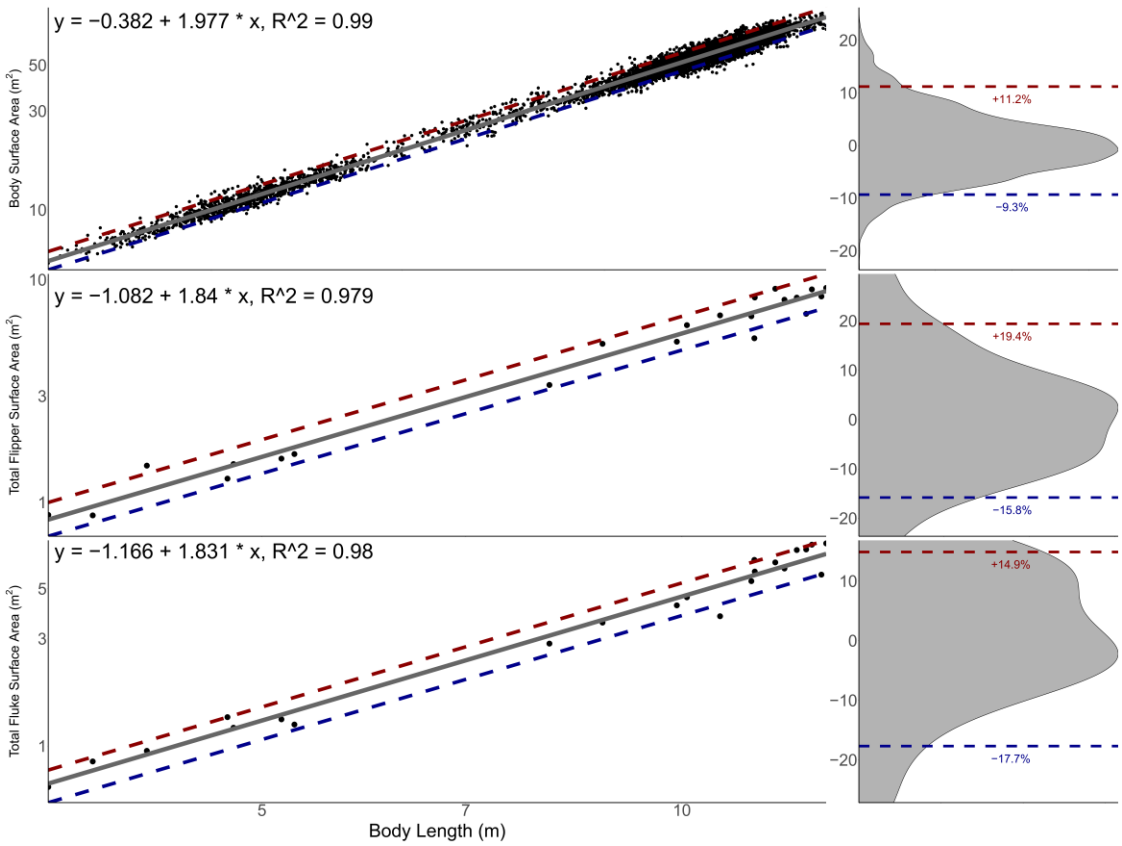


Figure S2: Morphometric relationships between body length and surface area estimates for A) the body, B) the flippers, and C) the flukes. Body values are taken from our broader dataset, while extremity values are taken from our subset of UAS data with clear flippers and flukes at the surface. Density plots show the residuals for each relationship (D-F), with the blue and red dotted lines denoting the 5% and 95% CI of each dataset, respectively.

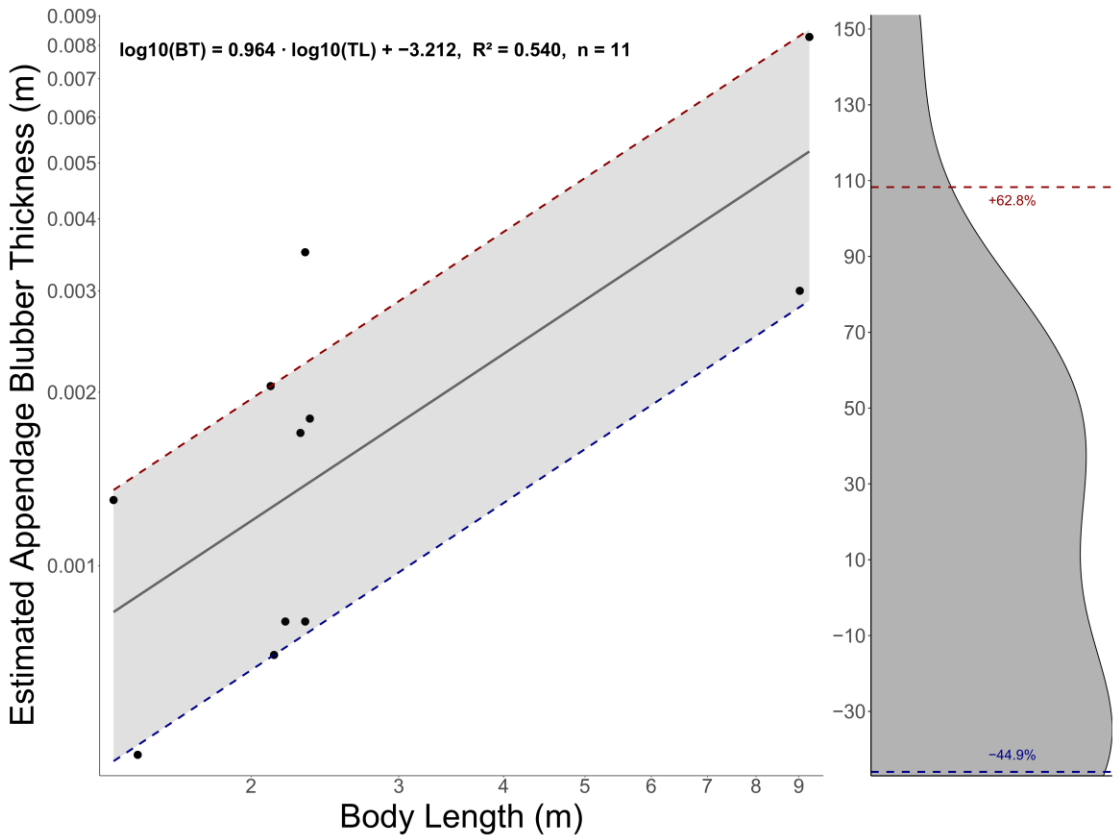


Figure S3: Morphometric relationships between body length and estimated appendage blubber thickness taken from our stranded animal dataset. Density plots show the residuals for each relationship (D-F), with the blue and red dotted lines denoting the 10% and 90% CI of each dataset, respectively.

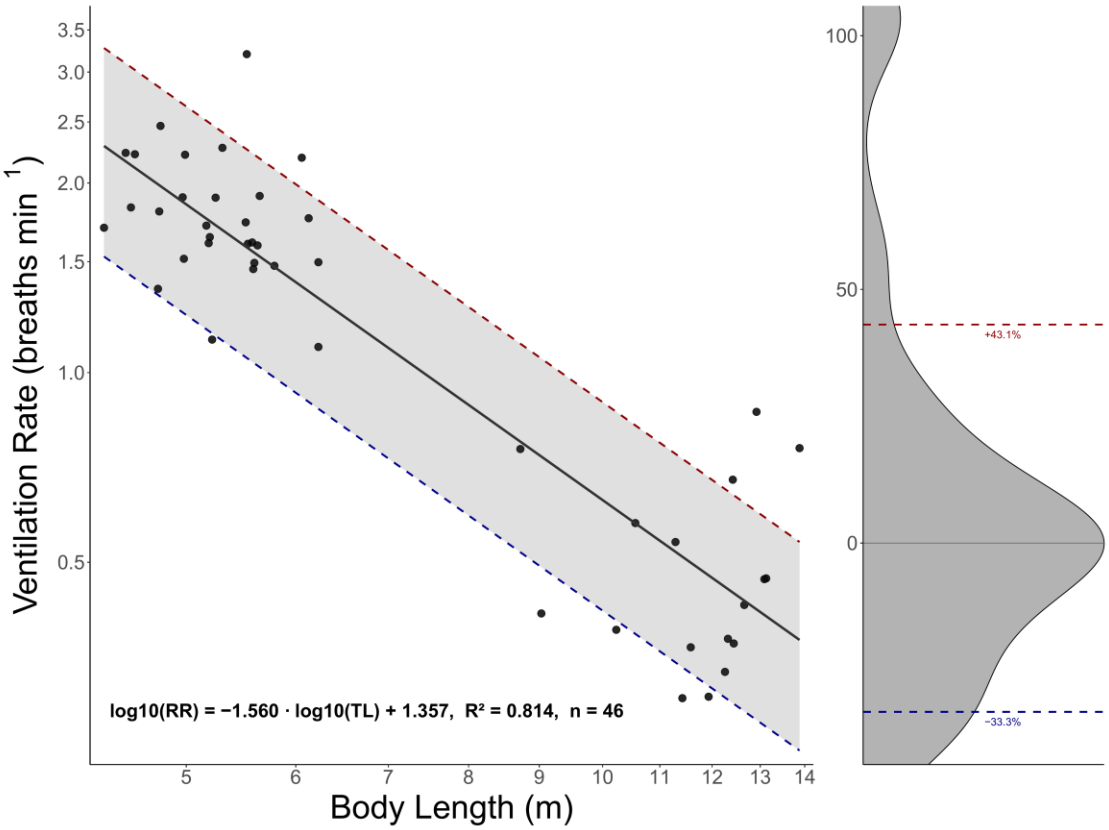


Figure S4: Log-transformed regression between body length and ventilation rate, taken from CATS tag data. The density plot shows the residuals for each individual, with the blue and red dotted lines denoting the 10% and 90% CI of each dataset, respectively.

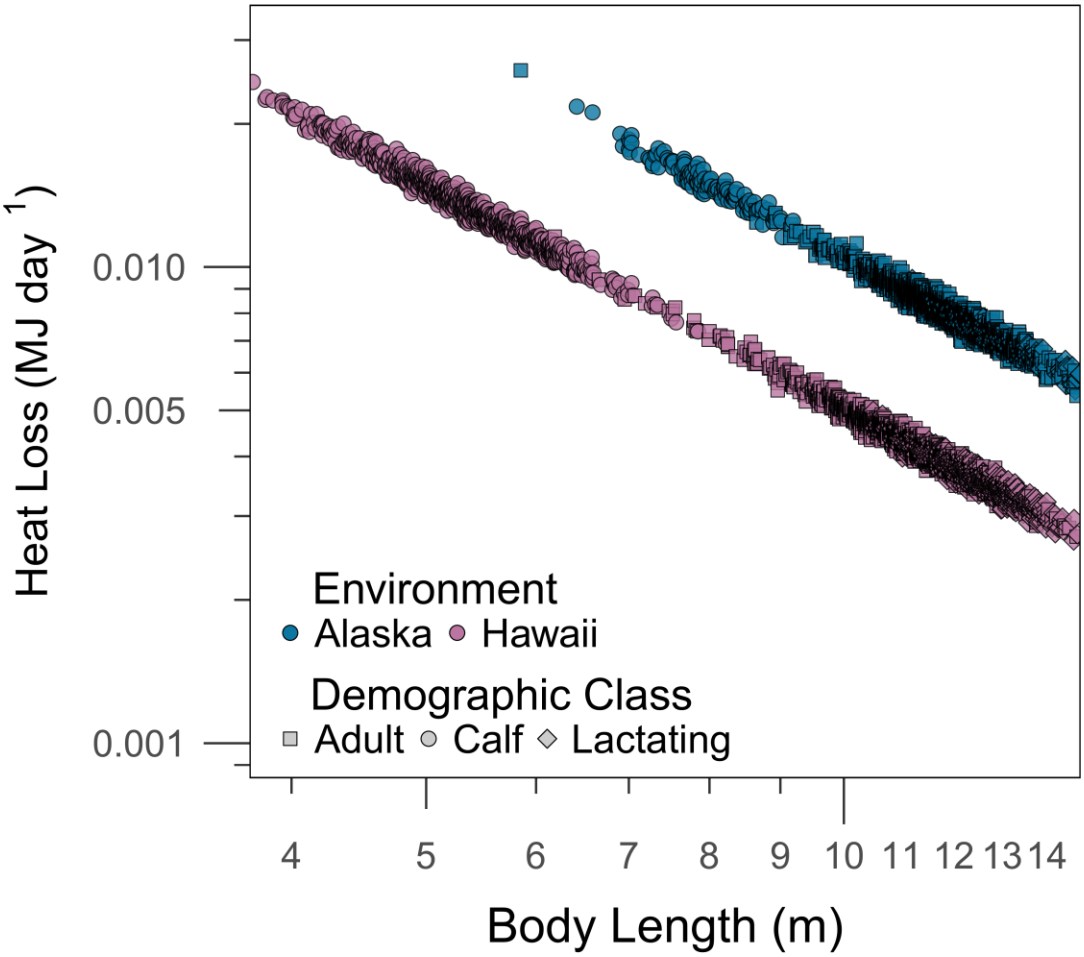


Figure S5: Ventilatory heat loss in Alaska (blue) and Hawai'i (pink).

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